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Tarea 10

atributos

R1

A	X	B	Y
7	2	6	11
3	4	9	15
10	7	2	4
1	12	2	11

4 registros

atributos

R2

B	W	D	Y	A	Z
2	5	6	11	1	30
4	7	8	4	7	8
9	10	11	28	5	12

3 registros

$$R1 X R2$$

$$R2 \bowtie R1$$

$$R1 \bowtie_{((R1.A > R2.Z \text{ or } R1.A >= R2.W) \text{ and } R1.Y = R2.Y)} R2$$

Tabla que incluya:

- atributos
- registros

Crear tablas:

```
CREATE TABLE R1 (
    A INT,
    X INT,
    B INT,
    Y INT
);
```

R1

```
CREATE TABLE R2 (
    B INT,
    W INT,
    D INT,
    Y INT,
    A INT,
    Z INT
);
```

R2

```
INSERT INTO R1 (A, X, B, Y) VALUES
(7, 2, 6, 11),
(3, 4, 9, 15),
(10, 7, 2, 4),
(1, 12, 2, 11);
```

Valores de R1

```
INSERT INTO R2 (B, W, D, Y, A, Z) VALUES
(2, 5, 6, 11, 1, 30),
(4, 7, 8, 4, 7, 8),
(9, 10, 11, 28, 5, 12);
```

Valores de R2

1) R1xR2

Atributos: 10 ya que 4 + 6 = 10 (R1 R2)

Registros: 12 ya que 4x3 = 12

R1.A	R1.X	R1.B	R1.Y	R2.B	R2.W	R2.D	R2.Y	R2.A	R2.Z
7	2	6	11	2	5	6	11	1	30
7	2	6	11	4	7	8	4	7	8
7	2	6	11	9	10	11	28	5	12
3	4	9	15	2	5	6	11	1	30
3	4	9	15	4	7	8	4	7	8
3	4	9	15	9	10	11	28	5	12
10	7	2	4	2	5	6	11	1	30
10	7	2	4	4	7	8	4	7	8
10	7	2	4	9	10	11	28	5	12
1	12	2	11	2	5	6	11	1	30
1	12	2	11	4	7	8	4	7	8
1	12	2	11	9	10	11	28	5	12

```
SELECT *
FROM R1
CROSS JOIN R2;
```

2) $R2 \bowtie R1$

Es join natural, se igualan los atributos comunes A, B, Y.

Se buscan registros donde

$R2.A = R1.A$

$R2.B = R1.B$

$R2.Y = R1.Y$

Solo coincide 1 registro

Atributos: 7

Registros: 1

B	W	D	Y	A	Z	X
2	5	6	11	1	30	12

```
SELECT *
FROM R2
NATURAL JOIN R1;
```

3) $R1 \bowtie_{((R1.A > R2.Z \text{ or } R1.A \geq R2.W) \text{ and } R1.Y = R2.Y)} R2$

Atributos: 10

Registros: 2

R1.A	R1.X	R1.B	R1.Y	R2.B	R2.W	R2.D	R2.Y	R2.A	R2.Z
7	2	6	11	2	5	6	11	1	30
10	7	2	4	4	7	8	4	7	8

```
SELECT *
FROM R1
JOIN R2
ON ((R1.A > R2.Z OR R1.A >= R2.W)
   AND R1.Y = R2.Y);
```

Primero se revisa $R1.Y = R2.Y$, despues si se cumple una de $\begin{cases} R1.A > R2.Z \\ \text{ó} \\ R1.A \geq R2.W \end{cases}$